MATERIAL AND DESIGN Come Together To Forge RELIABLE PCBs

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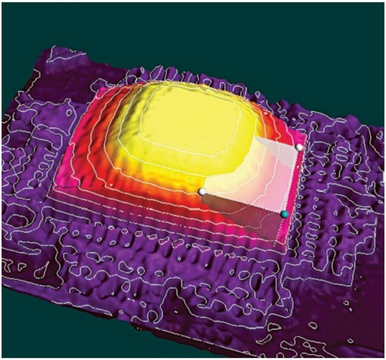
As we moved towards touch-screen phones with ultra-smart features, we also reduced the lifetime of our mobile phones. A case that comes to mind from personal experience is a phone bought in 2004 that was still functioning in 2014, and was decommissioned only when a different phone was bought.

The newer phones, however, do not look as if these are built to last. You eventually have to go for a change of device every couple of years or so. What does this say about the reliability of these devices?

Major threat to the reliability of a printed circuit board (PCB) in an electronic device comes from the heat produced and improper dissipation of heat. Most aspects of the PCB such as traces, solder masks, white-marking, laminates and plating are relatively reliable and rarely fail in use. However, copper barrels in the vias and holes can lead to long-term operational failures. Hence, reliability begins with the designing of the PCB itself.

Fig. 1: 3D visulaisation with heat mapping



EDA tools help with the design

The new generation of electronic design automation (EDA) tools have been a huge help with PCBs. Satyan Prakash Raj, development engineer, Toradex, says, "Latest EDA tools support both pre- and post-layout simulations, and help engineers analyse and fix issues during the design stage, much before getting assembled, and test it."

New EDA tools provide advanced routing features, design simulations including signal-integrity and post-layout automated design checks, PCB stack-up management and impedance calculations, which enable designers to build better circuit boards, lower design risks and bill of materials (BOM) management, among others.

Thermal management is a focus area in design. Problems brought in by heat can be sorted out by effectively mitigating the heat by proper selection of material sets chosen for building the PCB. This includes the chemicals involved in the process. Multilayered board lamination, circuit density and plated through-hole (PTH) technology are also factors to be considered while building the boards. And the most important of all, take time to get the design right. You never know when the next Samsung Galaxy Note 7 will happen.

Raj adds, "Latest tools come with integrated simulation and analysis tools for thermal analysis and impedance calculation." This could help in managing the problems caused due to heat generation at a later stage of the PCB build.

From a mechanical point of view, EDA tools allow you to generate and view the